Stack Data Structure

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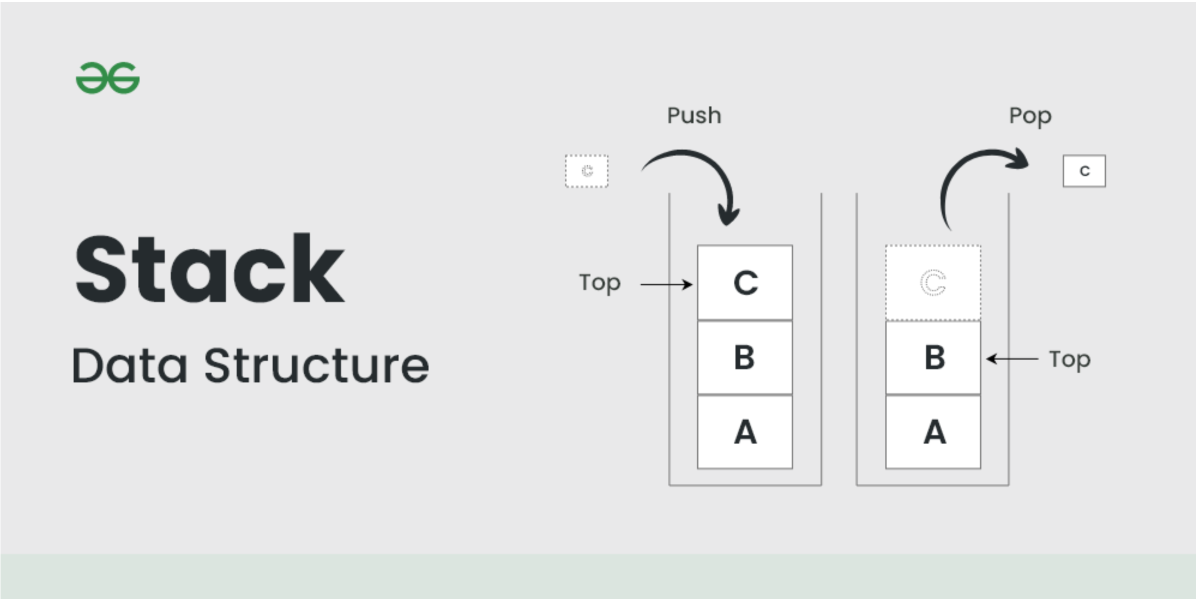
A **Stack** is a linear data structure that follows a particular order in which the operations are performed. The order may be **LIFO(Last In First Out)** or **FILO(First In Last Out)**. **LIFO** implies that the element that is inserted last, comes out first and **FILO** implies that the element that is inserted first, comes out last.

It behaves like a stack of plates, where the last plate added is the first one to be removed. **Think of it this way:**

- Pushing an element onto the stack is like adding a new plate on top.
- Popping an element removes the top plate from the stack.



Basics of Stack Data Structure

- Introduction to Stack
- Stack Array Implementation
- Stack Linked List Implementation
- Stack Implementation using Deque
- Applications of Stack

Implementations of Stack in Different Languages

- Stack in C++ STL

- Stack in Java
- Stack in Python
- Stack in C#
- Stack in JavaScript

Easy Problems on Stack Data Structures

- The Celebrity Problem
- Implement Queue using Stacks
- Implement two stacks in an array
- Implement Stack using Queues
- Stack using priority queue or heap
- Stack using single queue
- Infix to Postfix
- Prefix to Infix
- Prefix to Postfix
- Postfix to Prefix
- Postfix to Infix
- Infix To Prefix
- Check for balanced parentheses
- Arithmetic Expression Evalution
- Evaluation of Postfix Expression
- Reverse a stack using recursion
- Reverse Words
- Reverse a string using stack
- Reversing a Queue
- Reversing the first K of a Queue
- A Data Structure with O(1) Operations

Medium Problems on Stack Data Structures

- k Stacks in an Array
- Mergable Stack
- Previous Smaller Element
- Next Greater Element

- Stock Span Problem
- Buildings Facing Sun
- Next Smaller of next Greater in an array
- Next Greater Frequency Element
- Max product of indexes of greater on left and right
- Iterative Tower of Hanoi
- Sort a stack using a temporary stack
- Reverse a stack without using extra space in O(n)
- Delete middle of a stack
- Check if a queue can be sorted into another queue
- Check if an array is stack sortable
- Iterative Postorder Traversal | Set 1 (Using Two Stacks)
- Index of closing bracket for a given opening bracket
- Max Diff between nearest left and right smaller elements
- Delete consecutive same words in a sequence

Hard Problems on Stack Data Structures

- Largest Rectangular Area in a Histogram
- Sum of Max of all Subarrays
- Max of Mins of every window size
- Design a stack that supports getMin()
- Design a stack with max frequency operations
- Print next greater number of Q queries
- Length of the longest valid substring
- Iterative Postorder Traversal | Set 2 (Using One Stack)
- Print ancestors of a given binary tree node without recursion
- Expression contains redundant bracket or not
- Find if an expression has duplicate parenthesis
- Iterative method to find ancestors in a binary tree
- Stack Permutations
- Remove brackets from an algebraic string containing + and – operators
- Range Queries for Longest Correct Bracket Subsequence

Quick Links :

- ‘Practice Problems’ on Stack
- ‘Videos’ on Stack
- ‘Quizzes’ on Stack

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